IEEE COMPUTER SOCIETY
Kerala Chapter

IEEE

CS COMPILE

EDITION - 2

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With the rapid evolution of revolutionary technologies that aim at refining techniques in every sphere of life; it is without a doubt that developers strive their best to craft innovative ideas that can benefit mankind. Reflecting on IEEE's mission of fostering technological innovation and excellence for the benefit of humanity,

IEEE Computer Society Kerala Chapter 2024 is proud to carry on the legacy of providing imperative activities that can greatly benefit our members. Over the last year, the IEEE Computer Society Kerala Chapter has witnessed a robust number of members of the chapter actively being part of many of its events and has been a pivotal force in the successful completion of its activities. According to our statistics, the IEEE Computer Society Kerala Chapter has at present 29 graduate student members, 2 life seniors, 53 members, 35 senior members and 2,587 student members accounting for a total of 2,706 members. The chapter has about 45.08% of female members and 46.30 %. The statistics are a testimony to the steady progress of the IEEE Computer Society Kerala Chapter in being a significant advocate of innovative technology and creation, and we strive to broaden our horizons for the benefit of the members. By this timeline, IEEE Computer Society Kerala Chapter has conducted 6 events and 5 collaborative events that had witnessed fruitful conclusions and an active number of participants.

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EDITOR'S NOTE

RUGMINI SAI
CONTENT LEAD
IEEE Computer Society Kerala Chapter

Welcome to the second edition of "Compile," the official digital newsletter from IEEE Computer Society Kerala Chapter. Through this edition, we are delighted to introduce our readers to the fascinating world of Cloud Computing. Gaining immense recognition for its application and functionality, Cloud Computing has emerged as pivotal equipment in data management and organization. This edition of Compile delves into the rise of Cloud Computing and its profound impact as a revolutionary instrument in computing.

Through “Compile” you can explore articles that interpret the world of Cloud Computing through the perspectives of young minds and how it shapes the future of technology. These articles collect the different aspects of Cloud Computing providing insights on its evolution from its initial stages to its rapid growth as an indispensable tool for data organization and management. The Compile encompasses of the recent events of the IEEE Computer Society Kerala Chapter and the impact of each event. The success of these events is proof of the collaborative and vibrant nature of the Chapter, ranging from talk sessions that cover cutting-edge technologies to hackathons addressing real-life scenarios the design of each event reflects the core ideology of the Chapter. The snapshots of each event are a visual feast that echoes the energy encompassed in each event from the participants to the volunteers. The Compile also highlights the numerous collaborative events that symbolize the joined ventures of the Chapter with Student Branches across Kerala.

This edition of Compile takes this opportunity to acknowledge and felicitate the recipients of the REM Scholarship from the Kerala Section and the recipient of the 2024 R10 Volunteer Awards. Their achievements heighten the feats of success for the Chapter.
I sincerely express my gratitude towards the team behind the second edition of Compile, the content curators, designers, and editors whose tireless efforts led to the creation of this edition. The Compile is a testimony to the unwavering dedication and passion of the IEEE Computer Society Kerala Chapter team.

I hope the readers find this edition of Compile amusing and thought-provoking; invoking innovation and inspiration to the world of Cloud Computing. I am grateful for the support and guidance of our readers and I hope that this edition of Compile won't disappoint your expectations of IEEE CSKS.

Best Regards !

RUGMINI SAI
EDITOR-IN-CHIEF

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CLOUD COMPUTING:THE SKY IS JUST THE BEGINNING

In an era where technology evolves at breakneck speed,cloud computing has emerged as the backbone of modern digital transformation.From small startups to global tech giants, businesses are increasingly turning to the cloud to innovate, expand, and remain competitive. No longer merely a tool for enhancing business processes, cloud computing has become the platform that redefines what's possible in the digital age. The phrase "the sky's the limit" no longer applies when it comes to cloud computing—it's just the beginning of an endless frontier.

Traditionally, businesses relied on physical servers and on-premise infrastructure, which required substantial capital investment and ongoing maintenance. This outdated model often hampered growth and scalability. Cloud computing, however, has revolutionized this landscape. Today, companies can store, process, and access their data from virtually anywhere, at any time, without the exorbitant costs of maintaining physical servers. By adopting cloud-based solutions, businesses are empowered to scale operations quickly, reduce operational expenses, and embrace unparalleled agility.

The power of cloud computing lies in its flexibility and efficiency. At its core, the cloud provides on-demand access to computing power, storage, and resources that can be scaled up or down in real time. This allows businesses to respond dynamically to changing needs. For example, if a company experiences a sudden surge in demand, cloud infrastructure can expand instantly to accommodate that growth, ensuring smooth operations without downtime. When demand slows, resources can be scaled back, enabling cost savings. This pay-as-you-go model ensures companies only pay for the resources they use, making cloud computing an incredibly cost-effective solution.

Cloud computing offers other vital advantages, including shared resources and universal access. Instead of each company maintaining its own physical servers, cloud technology enables resource pooling across multiple users. This not only maximises efficiency but also enhances reliability. With cloud services, businesses have 24/7 access to their data and applications, whether on a laptop in the office or a smartphone on the go.

This "always-on" accessibility fosters collaboration, accelerates decision-making, and boosts productivity across teams

When selecting the right cloud solution, businesses have several models to consider. Public clouds, such as those offered by Amazon Web Services (AWS) and Microsoft Azure, are cost-effective, scalable options ideal for companies seeking flexibility without significant upfront investments. In contrast, private clouds offer more control and security, making them popular in industries like finance, healthcare, and government. Hybrid clouds combine the best of both worlds, merging the flexibility of public clouds with the security of private ones. This enables seamless, secure operations across platforms.

Cloud services are available in three primary forms: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). IaaS provides businesses with full control over their IT infrastructure without the hassle of managing physical hardware.PaaS, on the other hand, caters to developers by offering an environment where they can build, test, and deploy applications without worrying about the underlying infrastructure.With SaaS, businesses can quickly get up and running without complex installations or updates.

Beyond convenience and cost savings, cloud computing's greatest advantage is its unparalleled flexibility.As cloud computing continues to evolve, its potential appears limitless. While it has already transformed the way businesses operate, we are only scratching the surface of its capabilities. The cloud will continue to be the key to future-proofing businesses, offering the tools they need to innovate, scale, and outpace the competition.

In conclusion, the future of cloud computing is bright, and its role in shaping the business landscape is undeniable. The sky, once thought to be the limit, is no longer a boundary. Cloud computing has unlocked endless possibilities for businesses worldwide, and its journey is far from over. As companies explore new frontiers in the digital age, the cloud will remain the driving force behind innovation, growth, and success. Prepare to soar—because with cloud computing, the possibilities are truly endless.

THEJAL MARIA ABRAHAM

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CLOUD COMPUTING: AN INNNOVATIONAL IMPACT

Cloud computing has significantly transformed the way individuals and organizations manage and store data. It has moved beyond a mere technological trend to become an integral part of modern computing infrastructure. The shift to cloud services is not just about technology but also about changing the way we think about storage, computing power, and accessibility. This article explores the concept of cloud computing, its benefits, and its role in shaping the future of technology.

At its core, cloud computing refers to the delivery of computing services—including servers, storage, databases, networking, software, and analytics—over the internet ("the cloud"). Instead of relying on local hardware or data centers, businesses and individuals can use these resources remotely, paying for only what they use. This eliminates the need to invest heavily in expensive infrastructure and maintenance, making it a cost-effective solution for many. Furthermore, it offers flexibility, allowing users to scale their services up or down as per their requirements.

One of the most significant advantages of cloud computing is its accessibility. As long as you have an internet connection, you can access your data or applications from anywhere in the world. This has transformed how businesses operate, making remote work and collaboration more feasible than ever before. In the past, employees had to rely on local networks or physical storage devices to access their files. With cloud-based solutions, these barriers have been removed, leading to more efficient workflows and faster decision-making processes.

Another key aspect of cloud computing is its reliability and security. Many people initially expressed concerns about storing sensitive information on cloud platforms, worrying about data breaches or service outages. However, most cloud service providers offer robust security measures, including encryption, multi-factor authentication, and regular security updates.

Additionally, these providers often have backup systems in place to ensure that data is not lost in case of an outage. For businesses, this reliability is crucial as downtime can lead to significant financial losses. With cloud computing, companies can trust that their data and services will be available 24/7, offering peace of mind.

One of the most significant advantages of cloud computing is its accessibility. As long as you have an internet connection, you can access your data or applications from anywhere in the world. This has transformed how businesses operate, making remote work and collaboration more feasible than ever before. In the past, employees had to rely on local networks or physical storage devices to access their files. With cloud-based solutions, these barriers have been removed, leading to more efficient workflows and faster decision-making processes.

Cost-efficiency is another reason why cloud computing has gained widespread adoptions.In traditional setups, businesses needed to purchase expensive hardware, such as servers, and then maintain and upgrade them regularly. This not only required significant upfront investment but also ongoing costs for maintenance, electricity, and cooling.

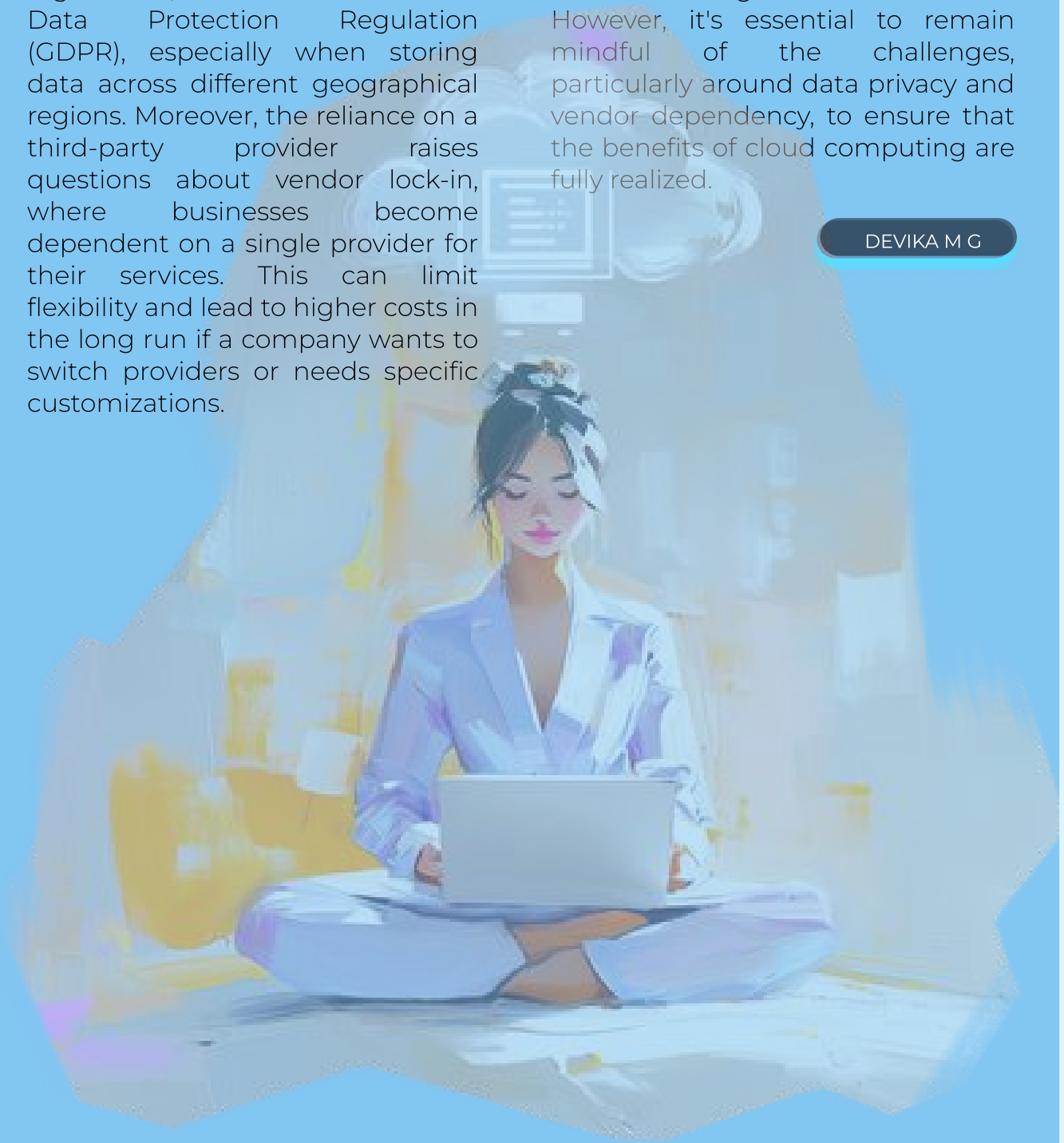
In contrast, with cloud services, businesses only pay for what they use. Whether it's computing power, storage, or bandwidth, cloud providers offer scalable solutions, which means businesses can start small and expand as needed without having to invest in new hardware. This model is especially beneficial for startups and small businesses that may not have the capital to invest in large-scale infrastructure.

Cloud computing has also played a pivotal role in innovation. The flexibility and scalability of cloud platforms enable developers and businesses to experiment with new ideas without worrying about infrastructure limitations. For instance, companies can deploy new applications, test them in real-time, and make improvements based on user feedback. This agility has shortened development cycles, allowing businesses to bring products and services to market faster.Additionally,cloud computing provides access to advanced technologies like artificial intelligence (AI) and machine learning (ML), which would otherwise be out of reach for many organizations due to the complexity and cost of implementing such technologies locally.

Despite its many benefits, cloud computing is not without challenges. One of the primary concerns is data privacy. While cloud service providers offer various security measures, the responsibility for protecting sensitive data still lies with the user. Companies need to ensure they are compliant with data protection regulations, such as the General Data Protection Regulation (GDPR), especially when storing data across different geographical regions. Moreover, the reliance on a third-party provider raises questions about vendor lock-in, where businesses become dependent on a single provider for their services. This can limit flexibility and lead to higher costs in the long run if a company wants to switch providers or needs specific customizations.

In conclusion, cloud computing represents a significant shift in how we think about and use technology. Its flexibility, cost-efficiency, and accessibility have made it a valuable tool for both individuals and businesses. As technology continues to evolve, the role of cloud computing is likely to expand, driving innovation and shaping the future of digital transformation. However, it's essential to remain mindful of the challenges, particularly around data privacy and vendor dependency, to ensure that the benefits of cloud computing are fully realized.

DEVIKA M G



SPOTLIGHT

The second edition of Compile 2024 is a testimony to the successful execution of several remarkable events by the IEEE Computer Society Kerala Chapter. Let's make a brief tour of the key events covered during the span of this edition.

On July 19-20, 2024, the IEEE Computer Society Kerala Section, in collaboration with IEEE SB NIT and IEEE Young Professionals Kerala Section, hosted **Slash Key 3.0**, a 30-hour national-level hackathon at the National Institute of Technology (NIT) Calicut. With a prize pool of ₹50,000, Slash Key 3.0 drew participants from across India. The event was generously supported by sponsors Institution's Innovation Council at NIT Calicut, .xyz, and Axure.

The hackathon began with a session titled Hackathon 101 led by Mr. Gokul C M,CEO and Founder of GADS LLP, where he shared valuable insights about hackathons and motivated the participants. Key Mentors from The IEEE CS SYP Kerala Chapter provided crucial guidance throughout the event, including Ms. Jyothika Nithin, Ms. Nayana S, Mr. Jayakrishnan P U, and Mr. Gokul G K. A judging panel consisting of Mr. Gokul C M, Mr. Dilber P Shakir (Oracle India), and Mr. Vivek Thrivikraman (Trizlabz Pvt Ltd) assessed the teams' projects. The hackathon was divided into three phases: ideation, development, and final presentation. After 30 hours of coding and problem-solving, Team Zhagaram emerged victorious, winning ₹30,000, followed by Team Rockstars (₹15,000), and Team Mugglers (₹5,000).

Slash Key 3.0 proved to be a successful platform for networking, collaboration, and technological advancement, fostering innovation and learning among participants.

In July and August 2024, the IEEE Computer Society Kerala Chapter organized a series of **Membership Development sessions** aimed at promoting IEEE membership, enhancing community engagement, and driving professional growth within the IEEE Computer Society. These sessions featured distinguished speakers and industry professionals who provided participants with valuable insights into the advantages of IEEE membership, the opportunities for volunteering, and the various resources available to members.

The series began on July 26, 2024,with a session conducted at the College of Engineering Chengannur by Jyothika Nithin, Malabar Hub Coordinator for IEEE CS SYP Kerala. The session focused on the significance of IEEE Computer Society membership, highlighting the access it provides to prestigious publications like the IEEE Computer Society Digital Library, technical competitions like IEEE Xtreme, and career-building opportunities such as the Richard E. Merwin Scholarship. The engaging Q&A segment motivated students to actively participate in IEEE initiatives. Following this, on August 19, 2024, a second session was hosted at SSET via Google Meet, featuring Nayana S, Travancore Hub Coordinator of IEEE CS SYP Kerala and Associate Cloud Engineer at DADA. The session emphasized career advancement through networking, skill development, and leadership opportunities within the IEEE Computer Society. Participants were also introduced to volunteer roles and global events that enable personal and professional growth.

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On May 3, 2024, as part of the **Elever 4.0** series, the IEEE Computer Society Kerala Chapter launched its second task, Pixel Perfect: The UI/UX Quest. This competition was designed to enhance participants' UI/UX design skills by challenging them to create functional, cross-platform-compatible wireframes for web or mobile applications. The task attracted talented designers from various institutions, leading to a highly competitive environment.

After an intense contest, the winners were announced: Godwin Kunjumon from FISAT secured first place, followed by Aravind Ashokan from MEC in second place, and Adwaith Jayasankar from SCET in third place. The Elever 4.0 series concluded with Government Engineering College Barton Hill taking the top spot on the overall leaderboard, followed by Sahrdaya College of Engineering and Technology and the Federal Institute of Science and Technology.

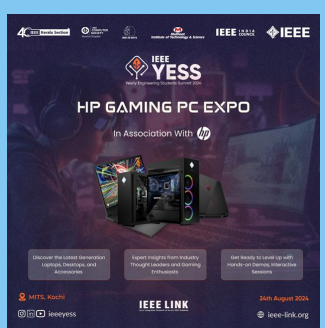
In August 2024, the IEEE Computer Society Kerala Chapter organized a series of **Distinguished Visitor Program (DVP)** sessions on intelligent systems, smart applications, and IoT. Led by experts, these sessions aimed to provide insights into emerging technologies while fostering collaboration among IEEE members.

On August 23, 2024, Prof. Mohamed Rawidean, IEEE R10 Regional Coordinator, presented "Smart Farming: Agriculture Transformation Using IoT, WSN, and AI" at CUSAT, highlighting smart technologies for sustainable agriculture. The next day, at Muthoot Institute of Technology, Kochi, Prof. Rawidean delivered a talk on "Advances in Intelligent Systems & Smart Applications," discussing the rapid progress of intelligent systems and smart city solutions.

These sessions inspired attendees to explore innovative ideas in IoT and smart applications, while fostering collaboration within the IEEE community.

Informatyka 4.0, in its current phase, has successfully conducted a total of 27 events. These events have focused on a wide range of topics, helping participants enhance their technical knowledge and practical skills. The diverse nature of the events allows students from different SBCs to collaborate within their designated hubs and gain exposure to various technical disciplines.

On August 24, 2024, the IEEE Computer Society Kerala Chapter, in collaboration with the IEEE Kerala Section, hosted the **HP Gaming PC Expo** at the Yearly Engineering Students Summit (IEEE YESS) 2024. Held at Muthoot Institute of Technology and Science, Kochi (MITS), the event was designed to give attendees an immersive experience of HP's state-of-the-art gaming technology. Participants had the opportunity to interact with HP gaming PCs, valued at over 5 lakh INR, through hands-on demonstrations at the Gaming Station.



The expo also featured a comprehensive PC Exhibit, where attendees explored the technical specifications and innovations behind HP's gaming systems. Expert sessions were conducted by HP's technical staff, covering topics such as the future of gaming hardware, the evolution of gaming experiences, and emerging trends in gaming technology. The event was a resounding success, providing a unique platform for gamers, tech enthusiasts, and students to delve into the world of high-performance gaming.

COLLABORATIVE EVENTS

The second edition of Compile 2024 is a testimony to the successful execution of several remarkable events by the IEEE Computer Society Kerala Chapter. Let's make a brief tour of the key collaborative events covered during the span of this edition.

Dev Summit 2024

Hosted by IEEE Computer Society SBC MITS in collaboration with IEEE CSKS and the MITS Cloud Tech Team on August 9-10, 2024, Dev Summit 2024 featured hands-on workshops on Web3, Golang, Rust, Flutter, and Generative AI. The event saw active participation from 101 attendees and included competitions like Blind Script and Capture the Flag, focusing on coding, problem-solving, and cybersecurity skills.

Final Round - Interview Tips and Tricks

Organized by IEEE Computer Society SBC MESCE in collaboration with IEEE CSKS, the Final Round session on 23rd August offered critical insights into technical and HR interview preparation. Mohammed Nihad, from UST Global, shared first-hand advice on effectively presenting skills and knowledge during interviews, concluding with an engaging Q&A session.

Empowering Kerala with Blockchain

As a community partner, IEEE CSKS supported the Empowering Kerala with Blockchain event organized by Web3Kerala, which featured prominent speakers including Chaal Pritam, Vish, Jubilant J. Kizhakkethottam, a professor and lead at SAINGIT's Blockchain Labs. This event focused on the transformative potential of blockchain technology in various sectors, fostering discussion and collaboration among enthusiasts and professionals in the field.

TechX INFINIA 2024

IEEE Sahrdaya SB, in collaboration with TechX, hosted TechX INFINIA 2024 from September 27th to 29th, 2024. On September 28th, a series of workshops were conducted covering topics such as machine learning, RPA, Flutter development, cybersecurity, and cloud computing. The event culminated in an intense 48-hour hackathon, offering participants an opportunity to showcase their skills and creativity.

HACKTHEFUTURE

IEEE SB TKMCE, in collaboration with IEEE CSKS, organized HACKTHEFUTURE, a 24-hour coding challenge on September 28-29, 2024, starting at 12 PM. This intense coding crucible tested participants' problem-solving and programming skills, offering a platform for innovation and collaboration in the tech space.

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RICHARD E. MERWIN SCHOLARSHIP

Gopika G Krishnan, from NSS College of Engineering, Palakkad, has received the honorable Richard E Merwin Scholarship for Spring 2024. This prestigious recognition highlights Gopika's dedication, exceptional contributions, and unwavering commitment to excellence within the IEEE community. Her relentless pursuit and passion for advancing technology have made her a deserving recipient of this esteemed scholarship.

Sreelakshmi A R ,Malabar Hub Coordinator SYP, has received the honorable Richard E. Merwin Scholarship for Spring 2024 . This prestigious award highlights Sreelakshmi's unwavering dedication, exceptional contributions, and commitment to excellence in her field. Her relentless efforts and passion for advancing technology have earned her this esteemed recognition, marking a significant achievement in her academic and professional journey.

Anna Ann Mathew ,Technical Coordinator of IEEE CSKS , from Mar Athanasius College of Engineering,Kothamangalam has received the honorable Richard E. Merwin Scholarship for Spring 2024 . This prestigious award highlights Sreelakshmi's unwavering dedication, exceptional contributions, and commitment to excellence in her field. Her relentless efforts and passion for advancing technology have earned her this esteemed recognition, marking a significant achievement in her academic and professional journey.

ACHIEVEMENTS

OUTSTANDING VOLUNTEER AWARD

Robin Francis Kochi HSR, the dedicated volunteer from Sahrdaya College of Engineering and Technology, has been honored with the prestigious 2024 IEEE Region 10 Outstanding Volunteer Award. This esteemed recognition celebrates Robin's remarkable contributions, leadership, and commitment to the IEEE community. His unwavering dedication to advancing technology and fostering collaboration within IEEE has set a shining example for others. Robin's exceptional service and passion for innovation have earned him this distinguished accolade, marking a significant milestone in his academic and professional journey.

IEEE IES GENERATIVE AI CHALLENGE 2024

Swastik H Nair, EIC of IEEE Computer Society Kerala Chapter, has achieved a remarkable feat by securing a position in the top 30 at the prestigious IEEE IES Generative AI Challenge 2024. This outstanding accomplishment highlights Swastik's expertise and innovation in the rapidly evolving field of AI. His success in this global competition reflects his dedication to advancing technology and his leadership within the IEEE community. This recognition marks a significant milestone in Swastik's journey of excellence in AI and beyond.

LANCE STAFFORD LARSON STUDENT AWARD

Mr. Justin Jacob, COO of Zimulate Consultech and Joint Secretary - PR, IEEE CS Kerala Chapter, has been honored with the prestigious Lance Stafford Larson Student Award 2023. This award celebrates Mr. Jacob's remarkable achievements and dedication in the field of technology and communication. His commitment and passion have earned him this esteemed award, marking a significant milestone in his professional journey.

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STUDENT SCHOLARSHIPS & AWARDS

RICHARD E. MERWIN SCHOLARSHIP

To recognize and reward active student volunteer leaders in student branches or chapters who show promise academically and professionally.

Deadline : 30th April & 30th September,2024

UPSILON PI EPSILON AWARD

For all graduate and undergraduate students who are IEEE Computer Society members and full-time students at an academic institution are eligible.

Deadline : October 31 of each year

LANCE STAFFORD LARSON BEST PAPER AWARD

For papers presented at a computer-related conference or in a publication. This award encourages students to develop excellence in their communication skills and to motivate students toward achievement in the field of Computer Science.

Deadline : October 31 of each year

IEEE CS MEMBERSHIP BENEFITS

- Magazines, Research, Journals, and Conference Proceedings: 18 complimentary articles per membership year from the most trusted source in computing.
- Spectrum Magazine: IEEE's flagship publication, available to those who belong to both IEEE and IEEE CS.
- IEEE Open Journal of the Computer Society: Save $350 on publishing papers with your IEEE Computer Society Membership.
- IEEE Computer Society Digital Library

  Access the premier collection of technical periodicals and conference papers. Included with student membership.

MEET THE *Team*

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